

HOMEWORK 14
INTRODUCTION TO COMPLEX ANALYSIS
(MATH 4230-40), SPRING 2022

SECTION 6.1 - ANALYTIC CONTINUATION AND ELEMENTARY RIEMANN SURFACES

1. Let $h(x)$ be a function of a real variable $x \in \mathbb{R}$. Suppose that $h(x) = \sum_{n=0}^{\infty} a_n x^n$, which converges for x in some interval around 0 $(-\eta, \eta)$, where $\eta > 0$. Prove that h is the restriction of some analytic function defined in a neighborhood of 0.
2. Prove the following result of Weierstrass:

Let $f(z) = \sum_{n=0}^{\infty} z^{n!}$. Then f cannot be analytically continued to any open set properly containing $A = \{z : |z| < 1\}$.

Hint: First consider $z = re^{1\pi i p/q}$ where $p, q \in \mathbb{Z}$.

3. Discuss the Riemann surface for $\sqrt[3]{z}$.
4. Consider the power series $\sum_{n=0}^{\infty} (-1)^n z^n$ defined in $|z| < 1$. To what domain in $\mathbb{C}$ can you analytically continue this function?